Final Project - Part1

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2026-03-05

1 - Read Data and EDA

```r
# Read Data
bike_raw <- read.csv("bike_sharing_daily.csv")

# Re-assign values of variables
bike_raw <- bike_raw %>%
  mutate(
    season     = factor(season,     levels = 1:4,
                        labels = c("Spring", "Summer", "Fall", "Winter")),
    mnth       = factor(mnth,       levels = 1:12),
    holiday    = factor(holiday,    levels = c(0,1), labels = c("No", "Yes")),
    weekday    = factor(weekday,    levels = 0:6,
                        labels = c("Sun","Mon","Tue","Wed","Thu","Fri","Sat")),
    workingday = factor(workingday, levels = c(0,1), labels = c("No", "Yes")),
    weathersit = factor(weathersit, levels = 1:4,
                        labels = c("Clear", "Mist/Cloudy", "Light Rain/Snow", "Heavy Rain/Snow"))
  )

# Filter for 2011
bike_2011 <- bike_raw %>% filter(yr == 0)
```

Categorical Variables

Season, month, holiday, weekday, and workingday are standard characteristics within a year. There is no reason to expect significant variation in their distributions. Thus, we will not print out summary tables and plots for these variables.

Weather Situation

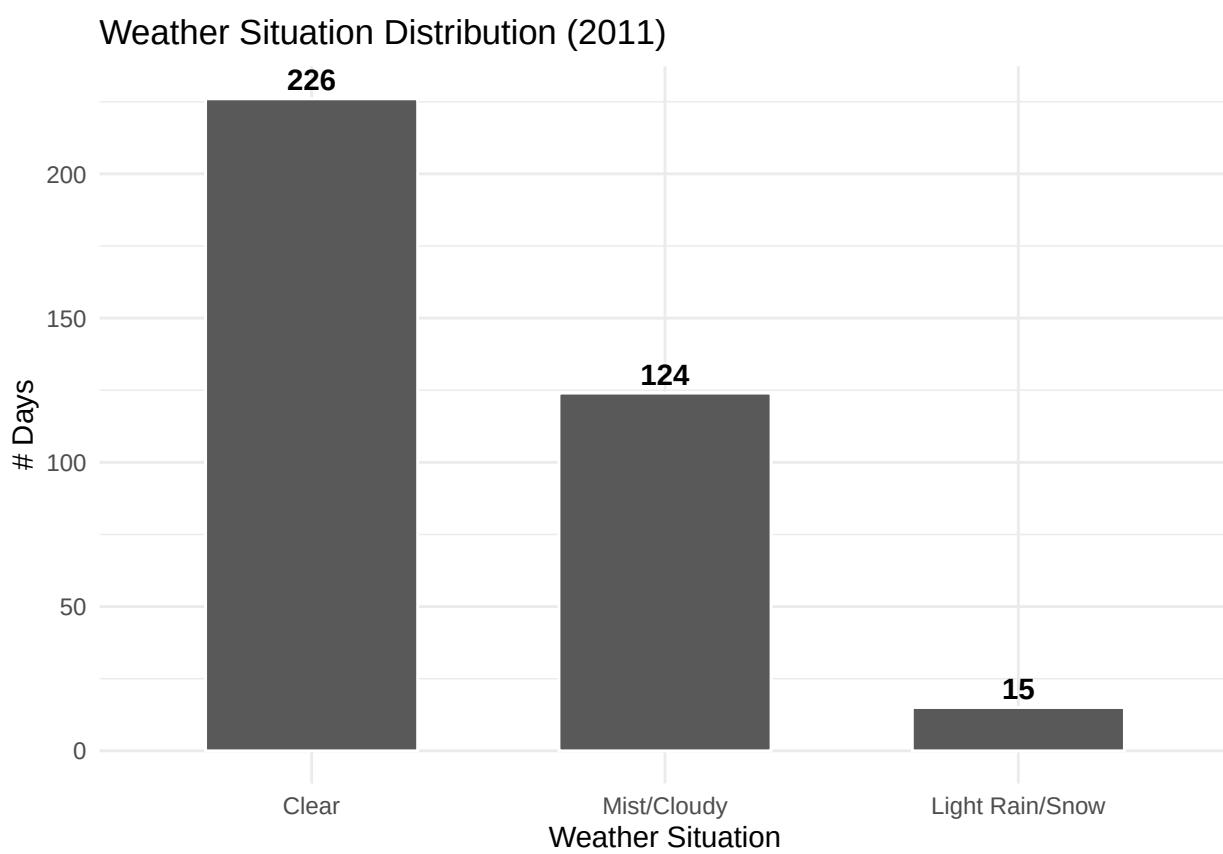
```r
table(bike_2011$weathersit)
```

```
##
##            Clear     Mist/Cloudy Light Rain/Snow Heavy Rain/Snow
##              226             124               15                0
```

```r
prop.table(table(bike_2011$weathersit))
```

```
##
##           Clear     Mist/Cloudy Light Rain/Snow Heavy Rain/Snow
##      0.61917808      0.33972603      0.04109589      0.00000000
```

```r
bike_2011 %>%
  count(weathersit) %>%
  ggplot(aes(x = weathersit, y = n)) +
  geom_col(width = 0.6, color = "white") +
  geom_text(aes(label = n), vjust = -0.4, fontface = "bold") +
  labs(title = "Weather Situation Distribution (2011)",
       x = "Weather Situation", y = "# Days") +
  theme_minimal()
```



Continuous Variables

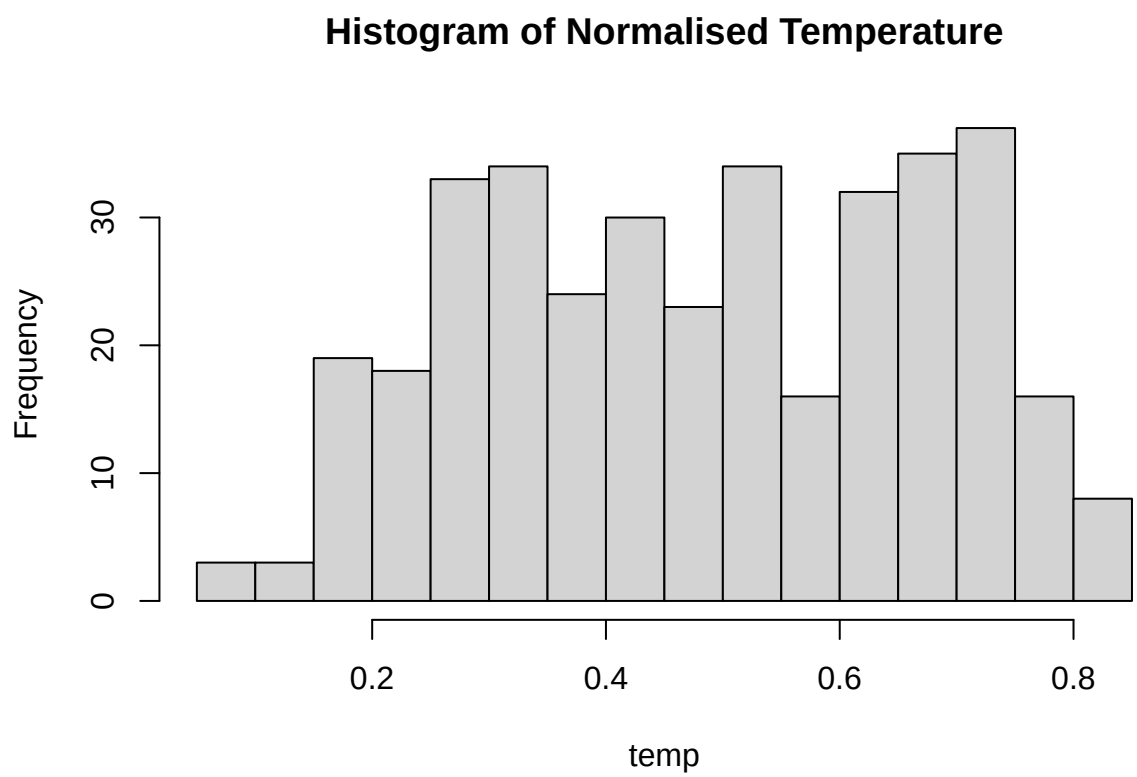
```r
cont_vars <- bike_2011 %>% select(temp, atemp, hum, windspeed, casual, registered, cnt)

# Numerical summary for all continuous variables
summary(cont_vars)
```

```
##       temp             atemp              hum           windspeed
##  Min.   :0.05913   Min.   :0.07907   Min.   :0.0000   Min.   :0.02239
##  1st Qu.:0.32500   1st Qu.:0.32195   1st Qu.:0.5383   1st Qu.:0.13558
```

```
##  Median :0.47917   Median :0.47285   Median :0.6475   Median :0.18690
##  Mean   :0.48666   Mean   :0.46684   Mean   :0.6437   Mean   :0.19140
##  3rd Qu.:0.65667   3rd Qu.:0.61238   3rd Qu.:0.7421   3rd Qu.:0.23508
##  Max.   :0.84917   Max.   :0.84090   Max.   :0.9725   Max.   :0.50746
##      casual         registered        cnt
##  Min.   :   9.0   Min.   : 416   Min.   : 431
##  1st Qu.: 222.0   1st Qu.:1730   1st Qu.:2132
##  Median : 614.0   Median :2915   Median :3740
##  Mean   : 677.4   Mean   :2728   Mean   :3406
##  3rd Qu.: 871.0   3rd Qu.:3632   3rd Qu.:4586
##  Max.   :3065.0   Max.   :4614   Max.   :6043
```

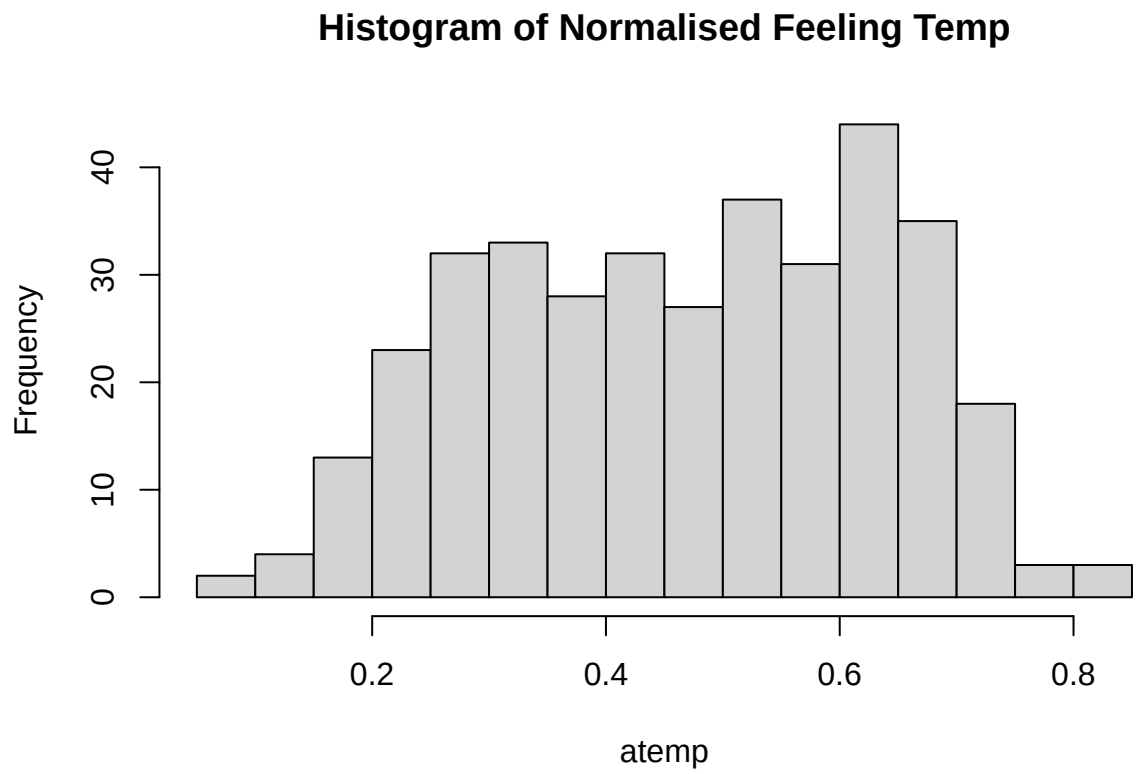
Temperature (`temp`)

```r
hist(bike_2011$temp, main = "Histogram of Normalised Temperature",
     xlab = "temp", breaks = 20)
```



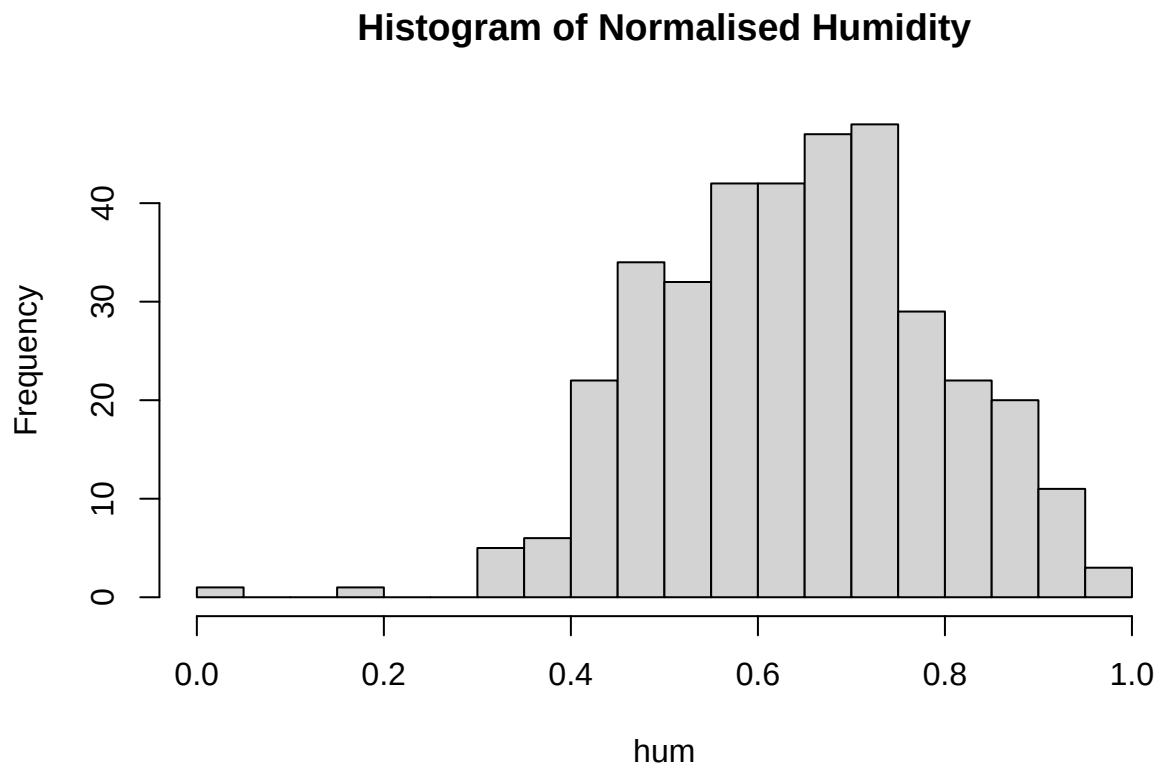
Feeling Temperature (`atemp`)

```r
hist(bike_2011$atemp, main = "Histogram of Normalised Feeling Temp",
     xlab = "atemp", breaks = 20)
```



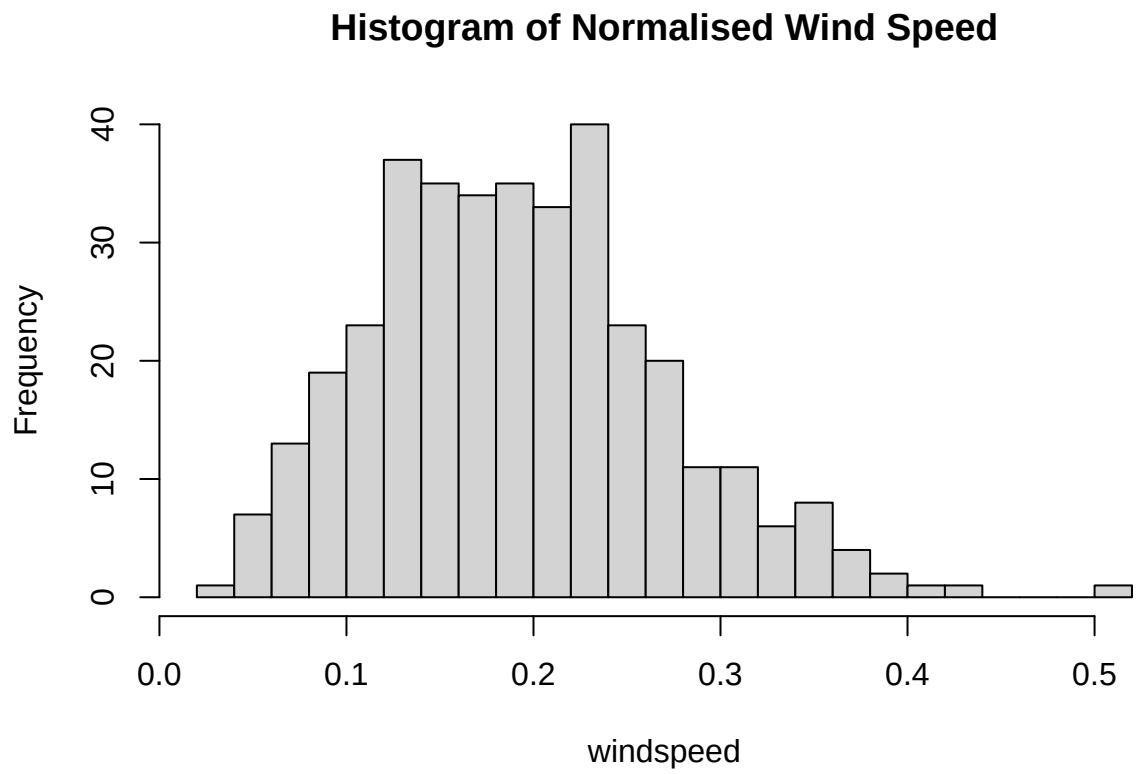
Humidity (`hum`)

```
hist(bike_2011$hum, main = "Histogram of Normalised Humidity",
     xlab = "hum", breaks = 20)
```



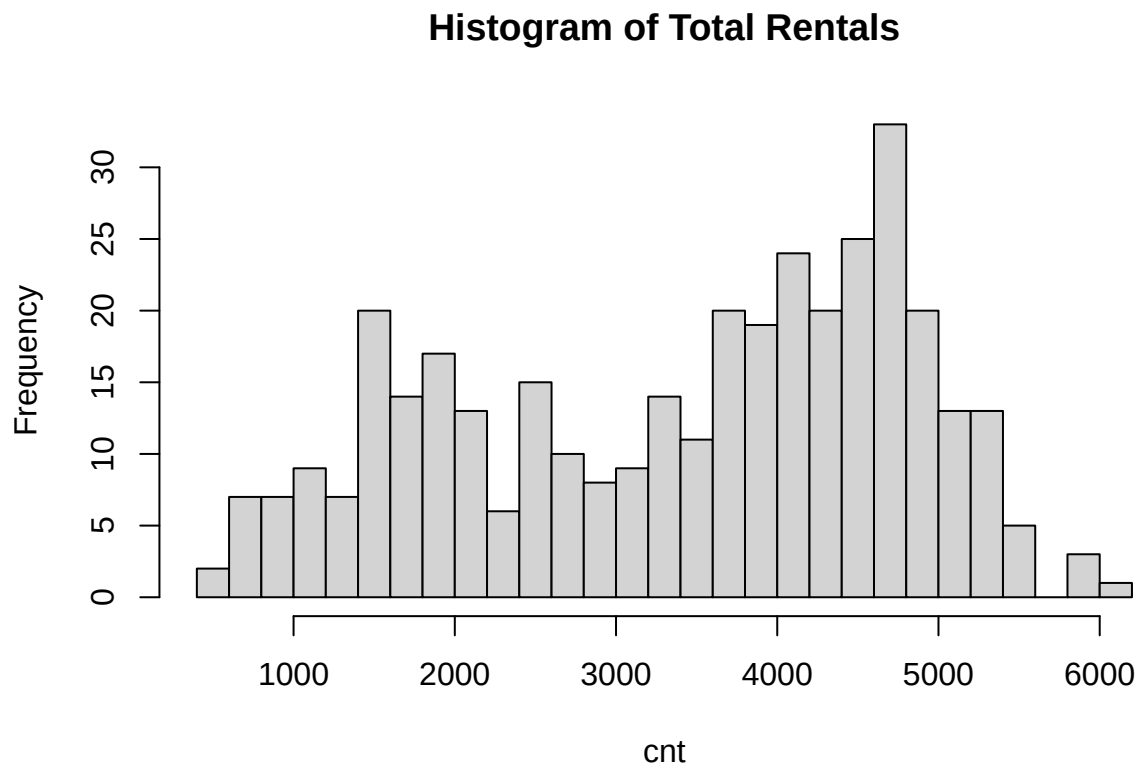
Wind Speed (`windspeed`)

```
hist(bike_2011$windspeed, main = "Histogram of Normalised Wind Speed",
     xlab = "windspeed", breaks = 20)
```



Total Rentals (`cnt`)

```
hist(bike_2011$cnt, main = "Histogram of Total Rentals",
     xlab = "cnt", breaks = 25)
```



2 - Fit linear regression models predicting `cnt` using each of the variables separately.

cnt ~ season

```
model_season <- lm(cnt ~ season, data = bike_2011)
summary(model_season)
```

```
##
## Call:
## lm(formula = cnt ~ season, data = bike_2011)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3349.4  -504.7    96.8   593.6  2029.8
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   1666.67      95.43   17.46   <2e-16 ***
## seasonSummer  2108.51     134.23   15.71   <2e-16 ***
## seasonFall    2797.70     133.52   20.95   <2e-16 ***
## seasonWinter  1997.79     135.34   14.76   <2e-16 ***
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 905.4 on 361 degrees of freedom
## Multiple R-squared:  0.5724, Adjusted R-squared:  0.5688
## F-statistic: 161.1 on 3 and 361 DF,  p-value: < 2.2e-16
```

The model explains about 57% of the variance in total rentals. The coefficients indicate that, compared to Spring, summer has 2108 more rentals, fall has 2797 more rentals, and winter has 1997 more rentals on average. Overall, season is a good predictor of total rentals.

cnt ~ mnth

```r
model_mnth <- lm(cnt ~ mnth, data = bike_2011)
summary(model_mnth)
```

```
##
## Call:
## lm(formula = cnt ~ mnth, data = bike_2011)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3357.2  -315.2    66.0   479.7  2149.7
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   1231.9      137.6   8.950  < 2e-16 ***
## mnth2          490.1      199.8   2.453   0.0147 *
## mnth3          834.1      194.7   4.285 2.36e-05 ***
## mnth4         1930.4      196.3   9.835  < 2e-16 ***
## mnth5         3149.4      194.7  16.179  < 2e-16 ***
## mnth6         3551.8      196.3  18.096  < 2e-16 ***
## mnth7         3327.5      194.7  17.094  < 2e-16 ***
## mnth8         3177.5      194.7  16.323  < 2e-16 ***
## mnth9         3015.4      196.3  15.363  < 2e-16 ***
## mnth10        2752.3      194.7  14.139  < 2e-16 ***
## mnth11        2173.7      196.3  11.075  < 2e-16 ***
## mnth12        1585.0      194.7   8.142 6.75e-15 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 766.4 on 353 degrees of freedom
## Multiple R-squared:  0.7004, Adjusted R-squared:  0.691
## F-statistic: 75.01 on 11 and 353 DF,  p-value: < 2.2e-16
```

The model has a better R2 than season, explaining about 70% of the variance in total rentals. Due to the strong corelation between month and season, the coefficients for month show a clear seasonal pattern. Rentals peak in the summer months (6, 7, 8).

cnt ~ holiday

```r
model_holiday <- lm(cnt ~ holiday, data = bike_2011)
summary(model_holiday)
```

```
##
## Call:
## lm(formula = cnt ~ holiday, data = bike_2011)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2986.1 -1285.1   329.9  1168.9  3040.8
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3417.13      73.19  46.689   <2e-16 ***
## holidayYes   -414.93     442.17  -0.938    0.349
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1379 on 363 degrees of freedom
## Multiple R-squared:  0.00242,    Adjusted R-squared:  -0.0003282
## F-statistic: 0.8806 on 1 and 363 DF,  p-value: 0.3487
```

The model has a very low R2 of 0.002, indicating that holiday status is not a good predictor of total rentals.

cnt ~ weekday

```r
model_weekday <- lm(cnt ~ weekday, data = bike_2011)
summary(model_weekday)
```

```
##
## Call:
## lm(formula = cnt ~ weekday, data = bike_2011)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2925.8 -1273.1   345.2  1147.7  2577.2
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3405.27     192.49  17.691   <2e-16 ***
## weekdayMon     60.52     272.22   0.222    0.824
## weekdayTue     62.77     272.22   0.231    0.818
## weekdayWed   -152.02     272.22  -0.558    0.577
## weekdayThu    -48.50     272.22  -0.178    0.859
## weekdayFri     94.85     272.22   0.348    0.728
## weekdaySat    -13.89     270.93  -0.051    0.959
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Residual standard error: 1388 on 358 degrees of freedom
## Multiple R-squared:  0.003175,   Adjusted R-squared:  -0.01353
## F-statistic: 0.1901 on 6 and 358 DF,  p-value: 0.9795
```

This model also has a low R2 of 0.003, suggesting that day of the week is a bad predictor of total rentals.

cnt ~ workingday

```r
model_workingday <- lm(cnt ~ workingday, data = bike_2011)
summary(model_workingday)
```

```
##
## Call:
## lm(formula = cnt ~ workingday, data = bike_2011)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2994.1 -1291.1   321.9  1164.9  2679.2
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)    3363.82     128.72  26.133   <2e-16 ***
## workingdayYes    61.24     155.53   0.394    0.694
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1380 on 363 degrees of freedom
## Multiple R-squared:  0.0004269,  Adjusted R-squared:  -0.002327
## F-statistic: 0.155 on 1 and 363 DF,  p-value: 0.694
```

This model's R2 is worse. At 0.0004, it indicates that whether it's a working day or not has no predictive power for total rentals.

cnt ~ weathersit

```r
model_weathersit <- lm(cnt ~ weathersit, data = bike_2011)
summary(model_weathersit)
```

```
##
## Call:
## lm(formula = cnt ~ weathersit, data = bike_2011)
##
## Residuals:
##    Min     1Q Median     3Q    Max
##  -3264  -1103    373   1032   2955
##
## Coefficients:
```

```
##                           Estimate Std. Error t value Pr(>|t|)
## (Intercept)                3694.99      86.74  42.598  < 2e-16 ***
## weathersitMist/Cloudy      -606.89     145.73  -4.164 3.90e-05 ***
## weathersitLight Rain/Snow -2020.85     347.69  -5.812 1.35e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1304 on 362 degrees of freedom
## Multiple R-squared:  0.1104, Adjusted R-squared:  0.1055
## F-statistic: 22.46 on 2 and 362 DF,  p-value: 6.363e-10
```

This model has an R2 of 0.11. The coefficients show that, compared to clear weather, adverse weather conditions (mist/cloudy, light rain/snow, heavy rain/snow) are associated with fewer rentals. However, the model's overall predictive power is still quite low.

cnt ~ temp

```r
model_temp <- lm(cnt ~ temp, data = bike_2011)
summary(model_temp)
```

```
##
## Call:
## lm(formula = cnt ~ temp, data = bike_2011)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3375.0  -501.2   -60.3   677.3  1985.7
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)    676.4      126.9   5.331 1.72e-07 ***
## temp          5608.3      243.0  23.082  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 878.9 on 363 degrees of freedom
## Multiple R-squared:  0.5948, Adjusted R-squared:  0.5937
## F-statistic: 532.8 on 1 and 363 DF,  p-value: < 2.2e-16
```

This model has an R2 of 0.59, indicating that temperature is a strong predictor of total rentals. The positive coefficient suggests that higher temperatures are associated with more rentals.

cnt ~ atemp

```r
model_atemp <- lm(cnt ~ atemp, data = bike_2011)
summary(model_atemp)
```

```
##
```

```
## Call:
## lm(formula = cnt ~ atemp, data = bike_2011)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3359.0  -536.9   -41.5   680.9  1924.1
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)    450.1      134.4    3.35 0.000893 ***
## atemp         6331.2      270.7   23.39  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 872 on 363 degrees of freedom
## Multiple R-squared:  0.6011, Adjusted R-squared:    0.6
## F-statistic:   547 on 1 and 363 DF,  p-value: < 2.2e-16
```

This model has an R2 of 0.6, slightly better than temp. The positive coefficient indicates that higher feeling temperatures are associated with more rentals.

cnt ~ hum

```r
model_hum <- lm(cnt ~ hum, data = bike_2011)
summary(model_hum)
```

```
##
## Call:
## lm(formula = cnt ~ hum, data = bike_2011)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2975.5 -1272.9   333.9  1180.9  2637.3
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3394.44     321.38  10.562   <2e-16 ***
## hum            17.59     486.51   0.036    0.971
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1381 on 363 degrees of freedom
## Multiple R-squared:  3.603e-06,  Adjusted R-squared:  -0.002751
## F-statistic: 0.001308 on 1 and 363 DF,  p-value: 0.9712
```

This model has a very low R2, the p-value for hum is also not significant. This suggests that humidity is not a good predictor of total rentals.

cnt ~ windspeed

```r
model_windspeed <- lm(cnt ~ windspeed, data = bike_2011)
summary(model_windspeed)
```

```
##
## Call:
## lm(formula = cnt ~ windspeed, data = bike_2011)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3361.4 -1071.1   275.1  1068.4  2601.7
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   4359.9      186.4  23.384  < 2e-16 ***
## windspeed    -4985.0      904.1  -5.514 6.67e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1326 on 363 degrees of freedom
## Multiple R-squared:  0.07728,    Adjusted R-squared:  0.07474
## F-statistic:  30.4 on 1 and 363 DF,  p-value: 6.673e-08
```

This model has a low R2 pf 0.07, indicating that wind speed is a weak predictor of total rentals.

3 - Fit multiple linear regression models with interactions.

Given that season, mnth, weathersit, temp, atemp are good predictors individually, we will explore interactions between these variables.
cnt ~ temp*weathersit

```r
model_temp_weathersit <- lm(cnt ~ temp * weathersit, data = bike_2011)
summary(model_temp_weathersit)
```

```
##
## Call:
## lm(formula = cnt ~ temp * weathersit, data = bike_2011)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3221.3  -544.7     3.9   625.4  1760.1
##
## Coefficients:
##                                Estimate Std. Error t value Pr(>|t|)
## (Intercept)                      1052.1      142.1   7.406 9.38e-13 ***
## temp                             5290.6      263.7  20.063  < 2e-16 ***
## weathersitMist/Cloudy            -733.7      254.6  -2.882  0.00419 **
## weathersitLight Rain/Snow       -1279.5      805.0  -1.589  0.11283
## temp:weathersitMist/Cloudy        618.0      499.2   1.238  0.21648
```

```
## temp:weathersitLight Rain/Snow   -976.5     1755.4  -0.556  0.57837
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 799.9 on 359 degrees of freedom
## Multiple R-squared:  0.668,  Adjusted R-squared:  0.6634
## F-statistic: 144.5 on 5 and 359 DF,  p-value: < 2.2e-16
```

This model has an R2 of 0.67, which is an improvement over the temp-only model. The interaction terms are not significant, suggesting that the effect of temperature on rentals does not differ significantly across weather situations. However, the main effect of temp and cloudy weather remain strong and significant.

cnt ~ temp*season

```r
model_temp_season <- lm(cnt ~ temp * season, data = bike_2011)
summary(model_temp_season)
```

```
##
## Call:
## lm(formula = cnt ~ temp * season, data = bike_2011)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3344.1  -307.4   120.4   438.9  1922.7
##
## Coefficients:
##                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)          638.3      237.1   2.691  0.00745 **
## temp                3759.8      818.0   4.596 5.97e-06 ***
## seasonSummer        -201.2      388.7  -0.518  0.60507
## seasonFall          3652.7      783.0   4.665 4.37e-06 ***
## seasonWinter        1349.2      401.4   3.361  0.00086 ***
## temp:seasonSummer   2488.4      990.0   2.513  0.01240 *
## temp:seasonFall    -3512.5     1337.8  -2.626  0.00902 **
## temp:seasonWinter    172.8     1100.6   0.157  0.87531
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 745.7 on 357 degrees of freedom
## Multiple R-squared:  0.7131, Adjusted R-squared:  0.7075
## F-statistic: 126.8 on 7 and 357 DF,  p-value: < 2.2e-16
```

This model has an R2 of 0.71, which is better than both the temp-only and season-only models. The interaction terms are significant, indicating that the effect of temperature on rentals varies by season.

cnt ~ month*weathersit

```r
model_month_weathersit <- lm(cnt ~ mnth * weathersit, data = bike_2011)
summary(model_month_weathersit)
```

```
##
## Call:
## lm(formula = cnt ~ mnth * weathersit, data = bike_2011)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2538.00  -340.22    42.93   376.17  1569.71
##
## Coefficients: (5 not defined because of singularities)
##                                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)                       1235.53     150.50   8.209 4.91e-15 ***
## mnth2                              616.58     207.17   2.976 0.003131 **
## mnth3                             1099.94     212.85   5.168 4.08e-07 ***
## mnth4                             2506.76     223.96  11.193  < 2e-16 ***
## mnth5                             3374.94     207.17  16.291  < 2e-16 ***
## mnth6                             3710.33     202.46  18.327  < 2e-16 ***
## mnth7                             3313.54     192.13  17.246  < 2e-16 ***
## mnth8                             3436.95     198.48  17.316  < 2e-16 ***
## mnth9                             3628.65     240.12  15.112  < 2e-16 ***
## mnth10                            3373.30     209.87  16.073  < 2e-16 ***
## mnth11                            2402.02     204.71  11.734  < 2e-16 ***
## mnth12                            1554.87     204.71   7.596 3.11e-13 ***
## weathersitMist/Cloudy               47.47     228.63   0.208 0.835646
## weathersitLight Rain/Snow         -729.53     638.54  -1.143 0.254064
## mnth2:weathersitMist/Cloudy       -452.35     339.60  -1.332 0.183760
## mnth3:weathersitMist/Cloudy       -546.77     327.13  -1.671 0.095574 .
## mnth4:weathersitMist/Cloudy      -1010.89     324.73  -3.113 0.002012 **
## mnth5:weathersitMist/Cloudy       -639.44     323.47  -1.977 0.048880 *
## mnth6:weathersitMist/Cloudy       -587.88     336.74  -1.746 0.081768 .
## mnth7:weathersitMist/Cloudy         32.46     403.49   0.080 0.935938
## mnth8:weathersitMist/Cloudy      -1066.95     342.27  -3.117 0.001984 **
## mnth9:weathersitMist/Cloudy       -701.40     333.69  -2.102 0.036304 *
## mnth10:weathersitMist/Cloudy     -1147.80     334.92  -3.427 0.000686 ***
## mnth11:weathersitMist/Cloudy      -366.88     355.72  -1.031 0.303118
## mnth12:weathersitMist/Cloudy       297.24     338.10   0.879 0.379956
## mnth2:weathersitLight Rain/Snow        NA         NA      NA       NA
## mnth3:weathersitLight Rain/Snow   -451.94     789.25  -0.573 0.567288
## mnth4:weathersitLight Rain/Snow  -2217.76     905.71  -2.449 0.014854 *
## mnth5:weathersitLight Rain/Snow        NA         NA      NA       NA
## mnth6:weathersitLight Rain/Snow        NA         NA      NA       NA
## mnth7:weathersitLight Rain/Snow        NA         NA      NA       NA
## mnth8:weathersitLight Rain/Snow        NA         NA      NA       NA
## mnth9:weathersitLight Rain/Snow  -1951.99     755.71  -2.583 0.010220 *
## mnth10:weathersitLight Rain/Snow -2056.97     746.65  -2.755 0.006192 **
## mnth11:weathersitLight Rain/Snow  -845.02     745.21  -1.134 0.257637
## mnth12:weathersitLight Rain/Snow  -411.37     787.10  -0.523 0.601570
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 620.5 on 334 degrees of freedom
## Multiple R-squared:  0.8141, Adjusted R-squared:  0.7974
## F-statistic: 48.76 on 30 and 334 DF,  p-value: < 2.2e-16
```

This model produces the best R2 so far of 0.814. All of the months are significant predictors of rentals, and

some interaction terms are also significant, indicating that the effect of month on rentals varies by weather situation. This makes sense as the impact of weather on rentals is likely to be more pronounced in certain months (e.g cloudy weather may have higher effect in summer months when it's usually hot).